ISAAC SHEARD

Cambridge, MA • (763) 402-4963 • imsheard@mit.edu

Education

B.S. Aerospace Engineering **Expected May 2028**

Massachusetts Institute of Technology (MIT) - Cambridge, MA | GPA: 4.3/5.0

- **Relevant Coursework:** Advanced Machining; Materials & Structures; Signals & Systems; Aerospace Probability & Statistics; Autonomous Machines; Thermodynamics & Prop.; Fluid Dynamics; Controls; Aerodynamics (IP); Feedback Control Systems (IP); Dynamics (IP);

Work History

Robotics Engineering Intern

Skyrun AI – San Francisco, CA **May 2026-Aug 2026**

- Designed and built 2 autonomous ground rovers from scratch, owning mechanical, electrical, compute, sensor, and autonomy integration; took the second platform from chassis delivery to its first autonomous SLAM map in 9 days and field-proven autonomous navigation in 6 weeks
- Built the sim-to-real infrastructure that deployed autonomy across 2 mechanically different rovers and simulation, enabling 6 autonomous behaviors to transfer between platforms without rewriting the underlying autonomy logic
- Trusted with technical hiring, onboarding, and core robotics tooling, conducting interviews for master's-level engineering candidates, training new hires on the rover/autonomy stack, and owning core robotics tooling

Embedded Software Engineering Intern

AGCO **– Jackson, MN** **Jun 2024 - Aug 2024; Jun 2025 - Aug 2025**

- Led the design and implementation of a user interface and CAN signal routing for an integrated weather station system on the sprayer platforms
- Delivered up to 200x faster runtimes across four optimization projects and built calibration software for the RoGator sprayer tractor

Zero Robotics Researcher **Jan 2025 - Jan 2026**

MIT Media Lab, Space Enabled – Cambridge, MA

- Collaborated with NASA Ames to test Astrobee and refine the ISS deployment plan, ensuring readiness aboard the International Space Station (ISS)
- Served as instructor and Master of Ceremonies for the 2025 competition finals, supporting students through office hours and Piazza while stabilizing the IDE with unit tests and 24 bug fixes

NASA Space Grant Consortium Research Intern **Jun 2023 - Aug 2023**

University of Minnesota – Minneapolis, MN

- Perfected sun tracking payload by adding software features, improving target reliability, and enabling 360 degree tracking
- Programmed microcontrollers for stratospheric data collection and component control, integrating sensors for attitude, angular rate, magnetic field, acceleration, GPS, temperature, and pressure

Solid Propulsion Subteam Member & Executive IT Chair **Sep 2024 - Aug 2026**

MIT Rocket Team – Cambridge, MA

- Led FEA of a filament-wound CFRP motor case in Ansys/ACP- built the winding/laminate model, defined the orthotropic CFRP material, simulated internal-pressure and remote-displacement end loads, and correlated predictions to hydrostatic testing
- Machined nozzle carrier and forward closure, mixed/cast custom solid propellant, and designed grain geometry for low-altitude SRAD tests

Head RI3D Captain **Oct 2025 - Present**

FIRSTxMIT – Cambridge, MA

- Launched RI3D@MIT in its inaugural year, leading a ~25-student team through rapid CAD, design reviews, and software development; personally fronted ~$4,500 for time-critical parts and secured donated materials from local businesses, delivering a fully functional robot within 72 hours of kickoff

Captain, Programming Lead, and Driver of Team 2847 **Oct 2020 - Aug 2024**

FIRST Robotics – Fairmont, MN

- Led FRC Team 2847 as captain, programming lead, and robot driver, securing its first state berth, qualifying for world competition, and winning the Minnesota state championship
- Successfully organized $149,000 in fundraising through grant writing, community initiatives, and business searching, surpassing the program's total funds from the previous two decades

Selected Awards

- **Robotics:** Minnesota State Champion (2023), Minnesota All State Programmer (2024), Dean's List Semifinalist (2021-2022, 2022-2023), Autonomous Award (2024), Quality Award (2022)
- **MSHSL 'Triple A' (Athletics, Academics, Arts)** (2023-2024)

Skills

- **Programming:** Python, Java, C++, C, GNU Make, Arduino, GCODE, Qt, SVG (markup)
- **Engineering & Aerospace:** 3D Printing, Onshape, SolidWorks, Fusion360, Solid Edge, openMotor, openRocket, RASAero, Ansys
- **Development Tools:** Git (CLI & GUI), GitHub, CANalyzer, WinMerge, Command Shell, PowerShell, Agile
- Spanish-Speaking (Upper Intermediate)